

Emil Kim

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Education

Pusan National University

M.Sc. student in Artificial Intelligence
Graduate School Scholarship

Busan, South Korea
Mar 2025 – Present

Pusan National University

B.Sc. in Computer Science and Engineering
Samsung Dream Scholarship • Outstanding Graduate (Top 10 in Class)

Busan, South Korea
Sep 2020 – Jan 2025

Research Interests

Biosignal Analysis (EEG, ECG) • Multimodal Representation • Medical Image Analysis.

Research Experience

IC&ML Lab, Pusan National University

Research Assistant

Busan, South Korea
Sep 2024 – Present

- Developed EEG–fMRI cross-modal translation framework, improving reconstruction quality while reducing model parameters by $3.2\times$ and peak memory by $2.3\times$.
- Designed, trained and evaluated Transformer- and state-space-based architectures for multi-channel EEG representation learning, improving performance by 6.6% and reducing inference latency by 7.1% across three benchmark datasets.
- Developed a multi-scale CNN–Transformer architecture for EEG-based seizure detection, capturing local and long-range dependencies to achieve 99.55% accuracy on the CHB-MIT dataset with 238K parameters .
- Developed a CNN-based Parkinson’s disease classifier from SPECT images and applied saliency maps for model interpretability.

Teaching Experience

Pusan National University

Teaching Assistant — Introduction to Unix Programming (CB2001102-060)

Busan, South Korea
Mar 2026 – Jun 2026

- Designed programming assignments, led weekly practice sessions, and mentored undergraduate students.

Publications

Conference Proceedings

- [E. Kim](#) and J. K. Gahm, “MARBLE: Lightweight EEG-to-fMRI Translation via Mamba-Attention and ROI-Conditioned Decoding,” *MICCAI*, Strasbourg, France, Sep 2026. **[Paper]** **[Code]** **[Demo]**
- [E. Kim](#) and J. K. Gahm, “CGM-EEG: Cross-Gated Mamba for Spatio-Temporal EEG Representation Learning,” *IEEE ISBI* (Oral), London, UK, Apr 2026. **[Paper]** **[Code]**
- [E. Kim](#), Y. Kenes, S. D. Tran, and J. K. Gahm, “Multi-Scale Attention Network for EEG-based Seizure Detection,” *Korea Computer Congress (한국소프트웨어종합학술대회)*, Jeju, South Korea, Jul 2025. **[Paper]**

Journal Articles

- [E. Kim](#) and J. K. Gahm, “HSST-EEG: A Hybrid State-Space and Transformer Architecture for EEG Decoding,” *IEEE Access*, vol. 14, pp. 79188–79196, 2026. **[Paper]** **[Code]**
- [E. Kim](#), Y. Kenes, S. D. Tran, and J. K. Gahm, “Multi-Scale EEG Feature Extraction with Attention Mechanism for Seizure Detection,” *Journal of KIISE*, vol. 53, no. 6, pp. 497–506, 2026. **[Paper]**

Projects

Ziyatron AI

Almaty Kazakhstan

- Curated an EEG dataset from 92 pediatric subjects in collaboration with 2nd pediatric hospital in Almaty.
- Developed an LLM integrated clinical decision support platform combining an EEG based seizure detection model with GPT 4o.
- Presented this work at Digital Bridge 2025 in Astana, Kazakhstan.

Awards & Scholarships

1st Place, Three Minute Thesis (3MT), Pusan National University, South Korea

Aug 2026

- Presented EEG-fMRI translation research in a three-minute presentation for a non-specialist audience.

Best Paper Award, Physical AI International Forum (한국인공지능학회), Seoul, South Korea

Nov 2025

- Recognized for *HSST-EEG: A Hybrid State-Space and Transformer Architecture for EEG Decoding*.

Graduate School Scholarship, Pusan National University, Graduate School of AI

Mar 2025 – Present

Samsung Dream Scholarship, Samsung Dream Scholarship Foundation

Sep 2024 – Dec 2025

- Selected as one of two recipients per university for outstanding international students.

Skills & Interests

Programming: Python, C, C++, MATLAB

ML Frameworks: PyTorch, Hugging Face, Scikit-Learn

Tools: Git, Linux, Docker, CUDA, NumPy, Pandas

Language: Russian (Native), English (IELTS 8.0), Korean (TOPIK Level 5)